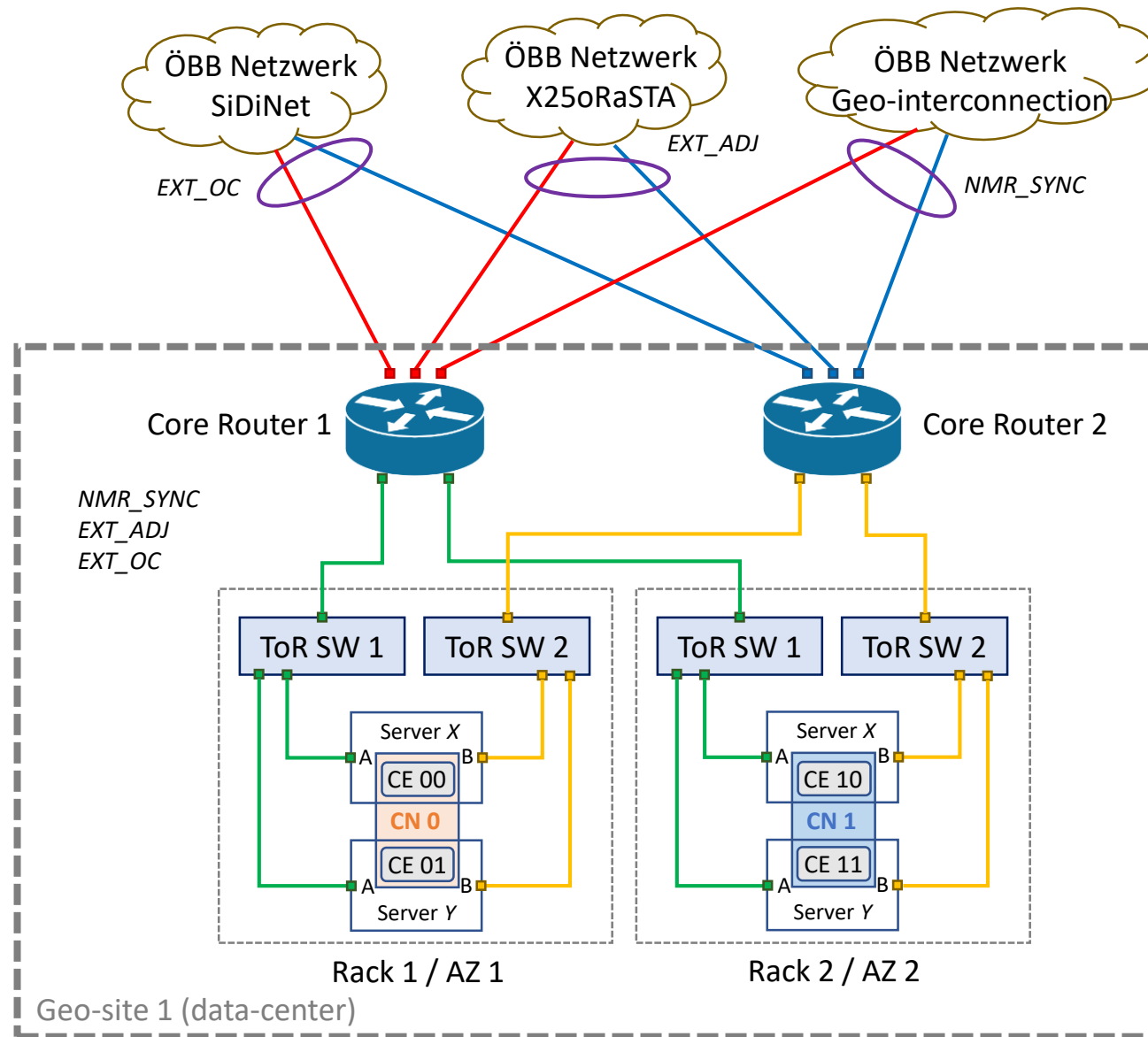


Network topology for the initial split-brain analysis in CG



Assumptions:

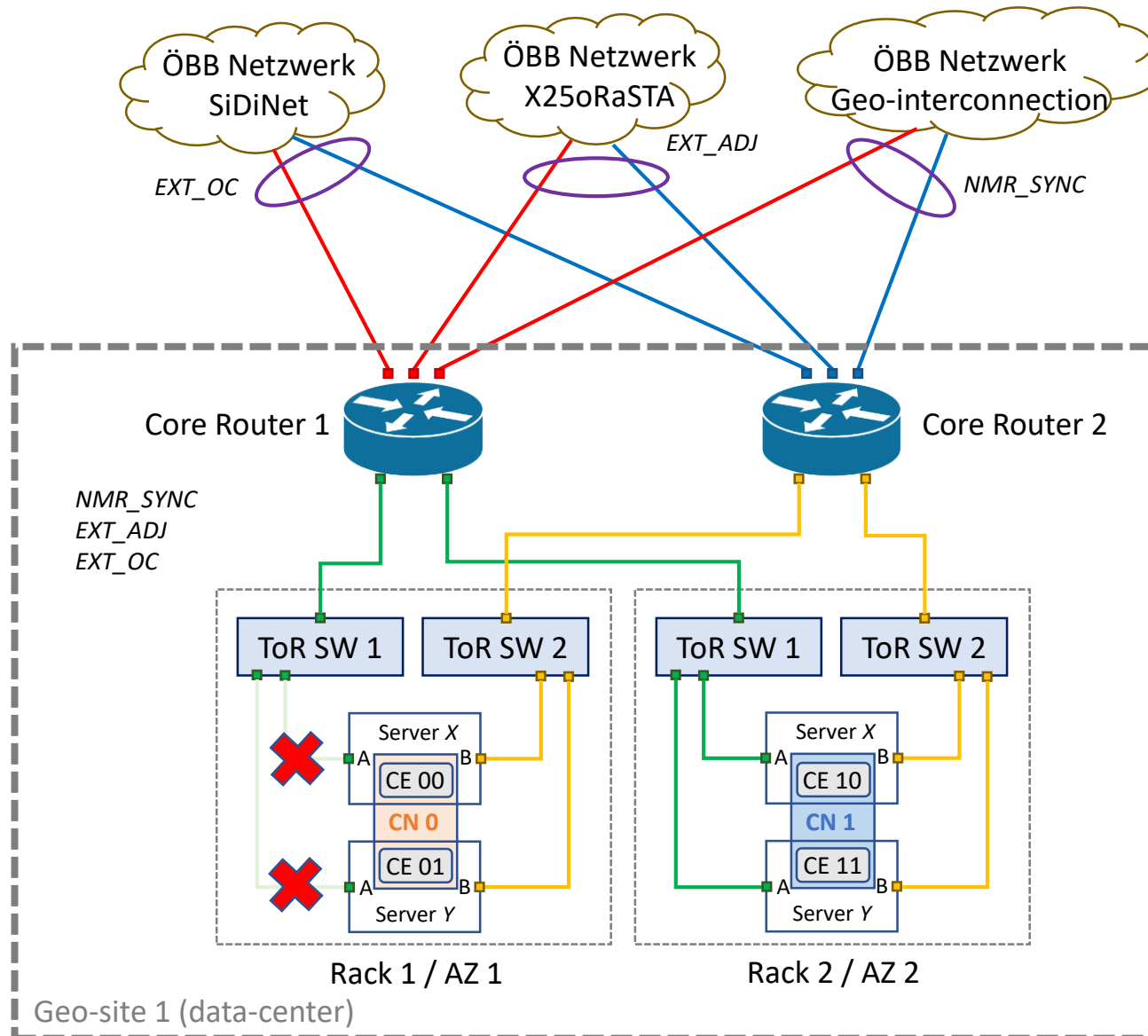
- ECMUX is not considered in this initial analysis
- One CN is inside one availability zone (rack)
- CEs of the same CN are deployed on different server blades
- There is at most one switch between two CEs of CN (SDD NMR and network requirements)
- "IPS" Network for other interfaces (SECLOG, SYSLOG, SECMNT, DLG ...) not relevant for this initial analysis, thus not shown and considered
- Bonded interface(s) for IPS control plane not relevant for this initial analysis, thus not shown and considered
- Single point of failure between communication points is excluded due to the network topology design
- Double failures are considered

Notation:

- *Italics* denotes VIL's virtual networks assigned to the physical network
- Ports A and B on the server blade are each from different network card, thus failure of one card affects only one channel and not both. The other 2 ports are not relevant for this initial analysis and model.

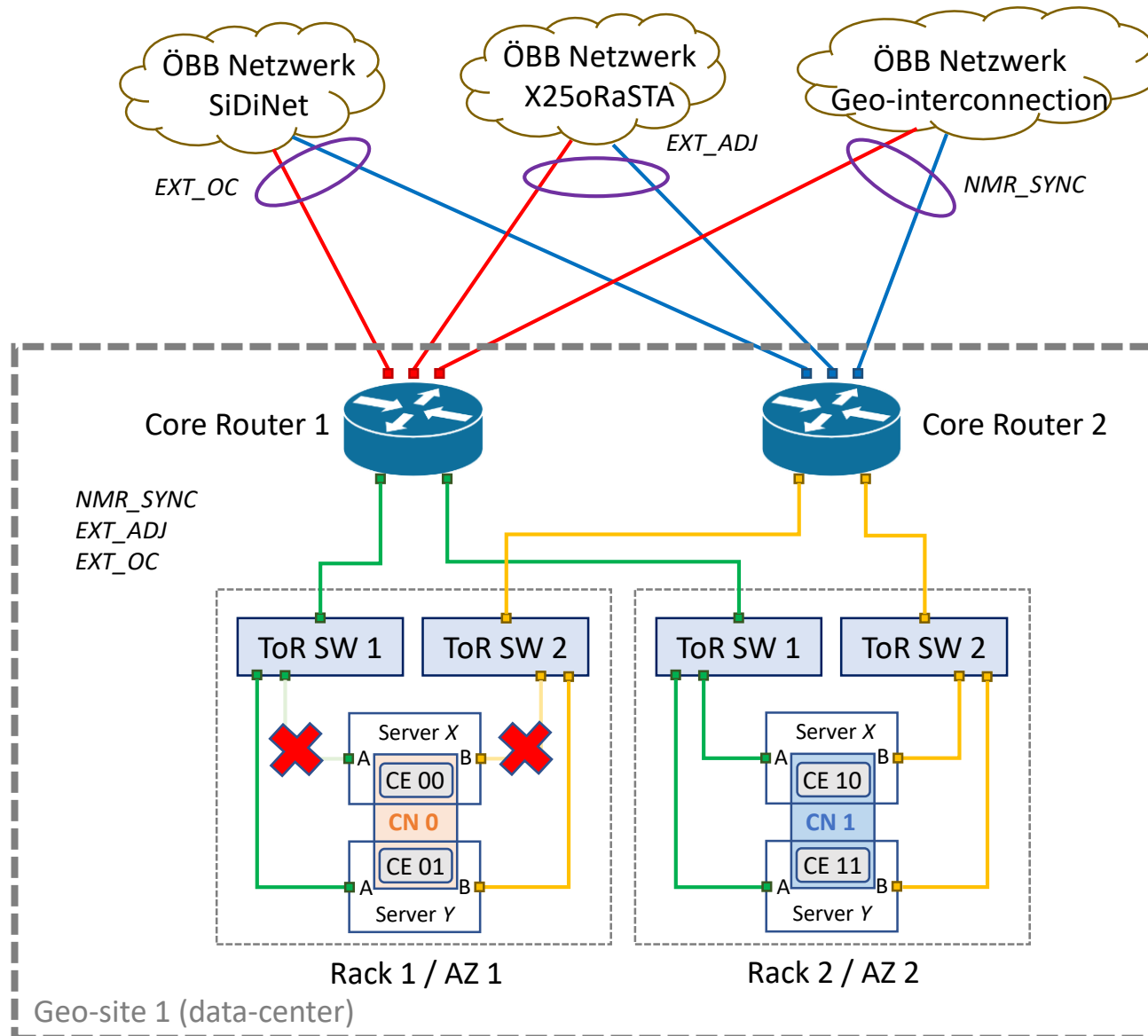
Failures inside rack (AZ)

Double failure inside rack



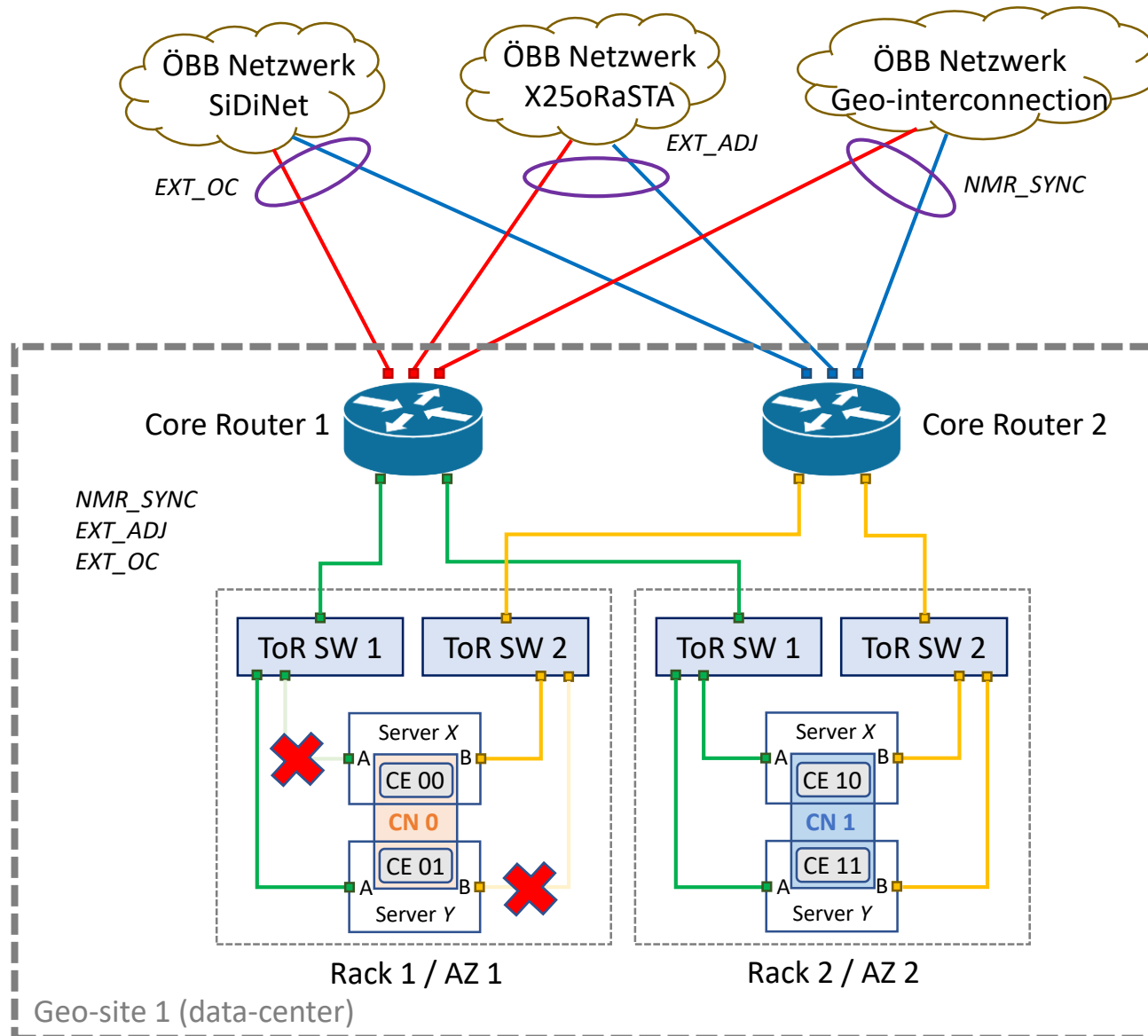
- **CN effect:** Only one physical channel but without effect on CN
- **CG effect:** CNO continues operation via “yellow” channel.
- **External networks:** No effect, full physical redundancy continues
- **No risk of split-brain**

Double failure inside rack



- **CN effect:** CN goes to the safe state. Connected CE will broadcast “shutdown” message to all CEs in CG (master token protocol)
- **CG effect:** If master – then new master is elected (master token protocol). If slave – no impact.
- **External networks:** No effect, full physical redundancy continues
- **No risk of split-brain**

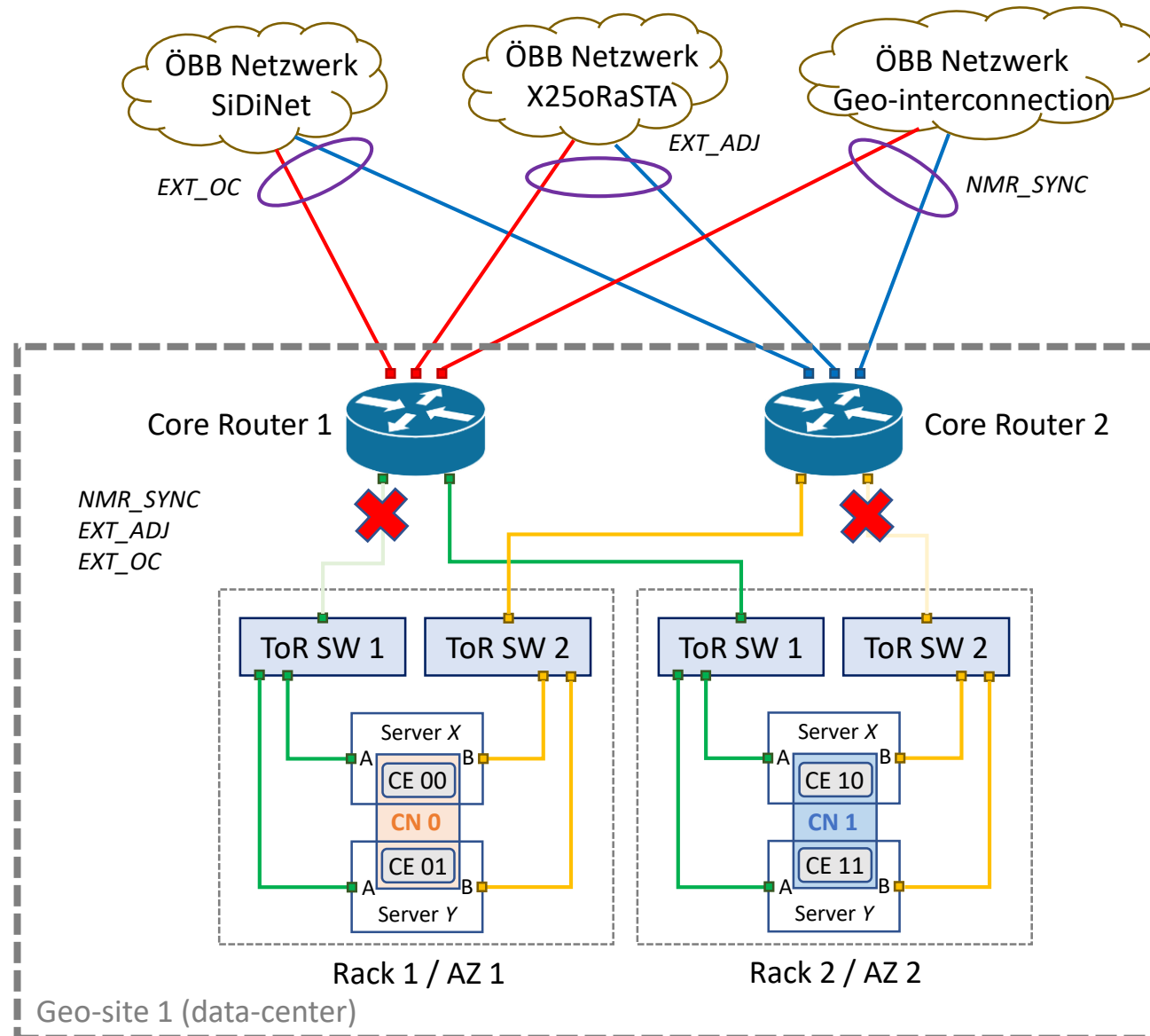
Double failure inside rack



- **CN effect:** CN goes to the safe state. Connected CE will broadcast “shutdown” message to all CEs in CG (master token protocol)
- **CG effect:** If master – then new master is elected (master token protocol). If slave – no impact.
- **External networks:** No effect, full physical redundancy continues
- **No risk of split-brain**

Failures between racks and core routers

Double failure between racks and core routers



- **CN:** Both CN0 and CN1 are operational
- **CG effect:**
 - If CN0 is master: CN1 can't see Master but from CN2, CN3 messages knows it exists. CN1 assumes it has network issues and will trigger PANIC (shutdown).
 - If CN0 is slave (master @ geo-site 2): No change, both CN0 and CN1 see the master
- **External networks:** No impact.
- **No risk of split-brain**

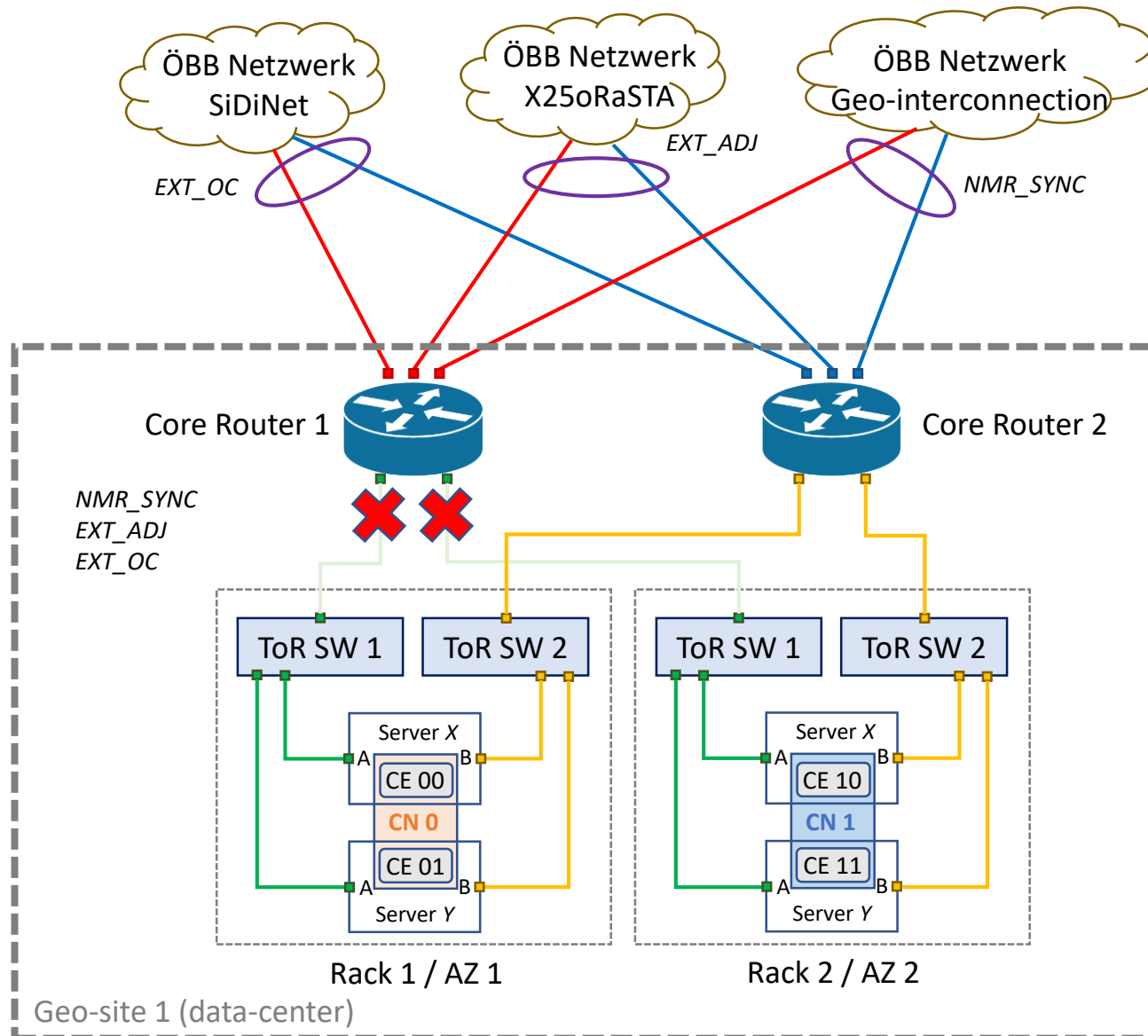
Created ticket for TAS:

<http://plf-support.at.gts/otrs/customer.pl?Action=CustomerTicketZoom;TicketNumber=2026021610000042>

Answers:

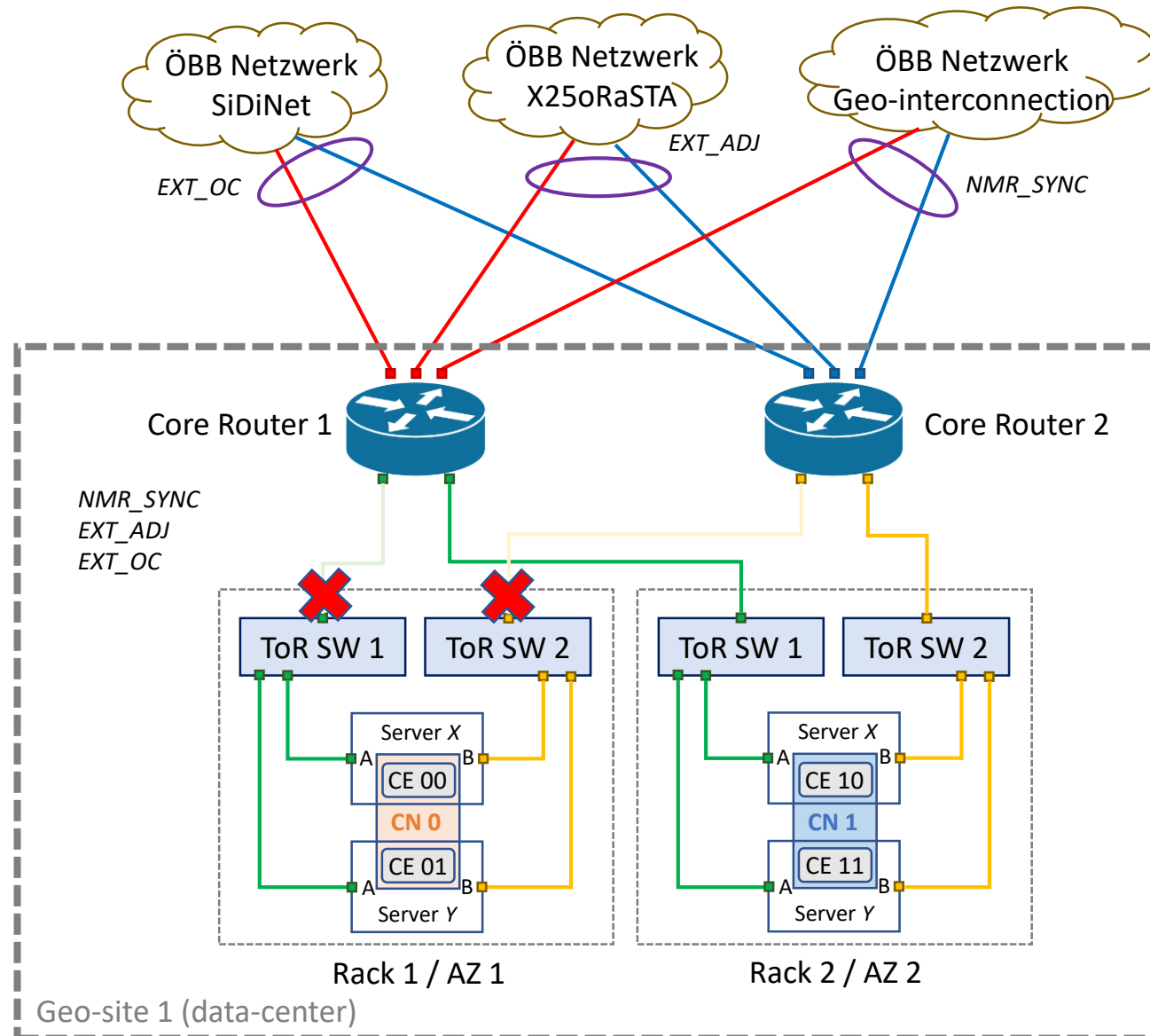
- "If the second site is still running and not having the Master role there, then the Slave on the first site will shutdown."
- Since CN1 cannot see CN0 directly it deduces from the messages that it gets from CN2 and CN3 that CN0 is still Master and that it, itself has a connection problem. NMR on CN1 then issues a PANIC.

Double failure between racks and core routers



- **CN effect:** No effect for both redundancies
- **CG effect:** No effect but both redundancies have only one physical channel.
- **External networks:** No effect but only one physical channel operational.
- **No risk of split-brain**

Double failure between racks and core routers



Master CN (on the left):

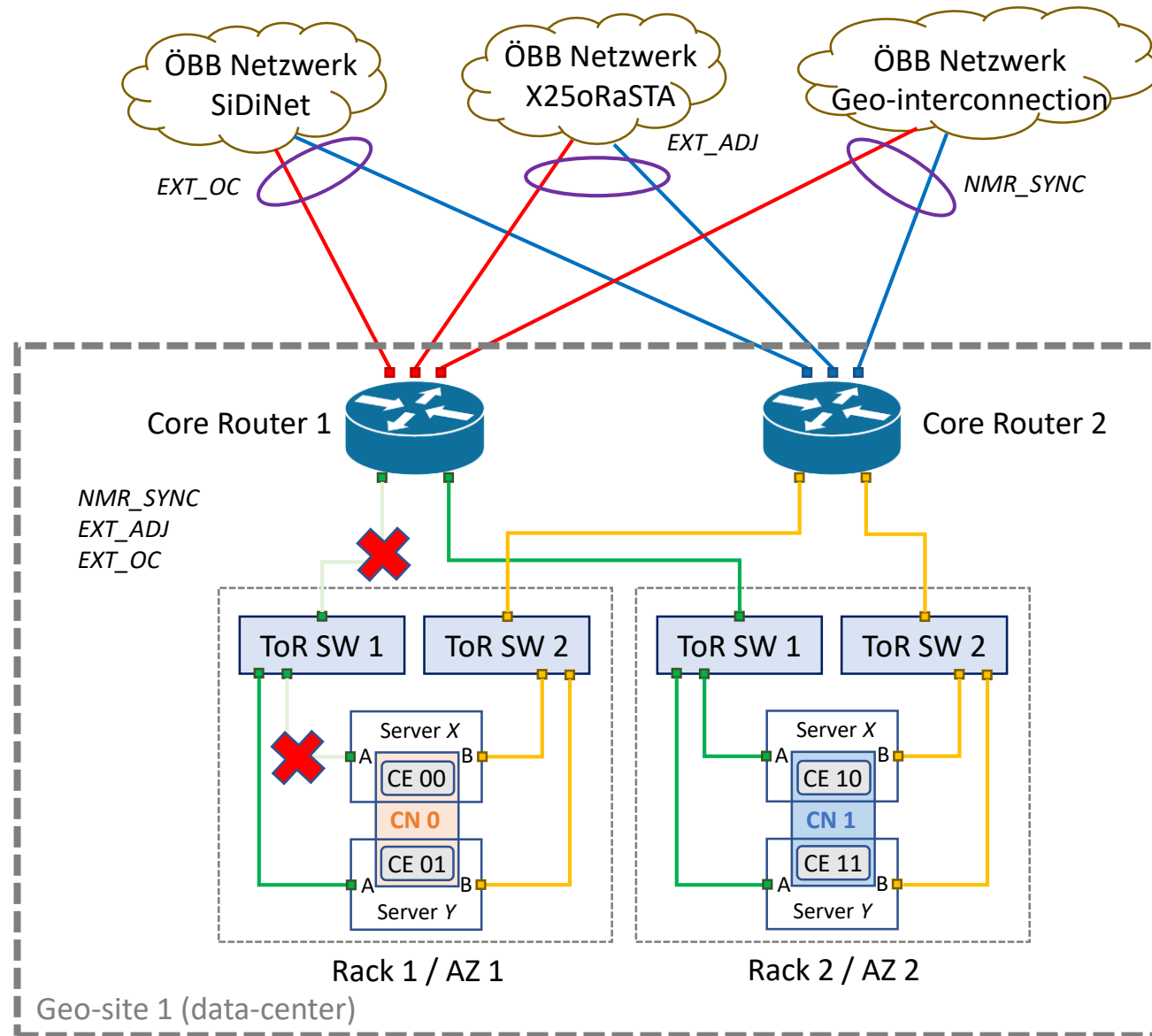
- **CN effect:** CN is operational but cannot send/receive any communication from CG and external networks. Remains master.
- **Effect on the rest CG:**
 - Master election process (MEP) starts
 - MTP does not elect master
 - Slaves will reset (master switch timeout). Without Observer will not be able to elect a new master at startup.
- **External networks:** No communication with Adjacent systems. EXT_OC – tbd
- **No risk of split brain**
 - No safety hazard because CN0 is fully isolated and cannot communicate with external networks
- **Availability issue – state is lost because slaves go to reset. When master is reintegrated – potential loss of state again.**

Slave CN:

- **CN effect:** CN is operational but cannot send/receive any communication from CG and external networks.
- **CG effect (default):**
 - The rest of CG continues normal operation
 - CN0 goes into master election process (MEP)
 - MTP does not elect master
 - Slave will reset. If configured for Observer, then will not elect master and continue in reset (if no MCI configured).
- **External networks:** No effect
- **No risk of split-brain**
 - CN0 can't decide to be the master and continues MEP until the connection is back

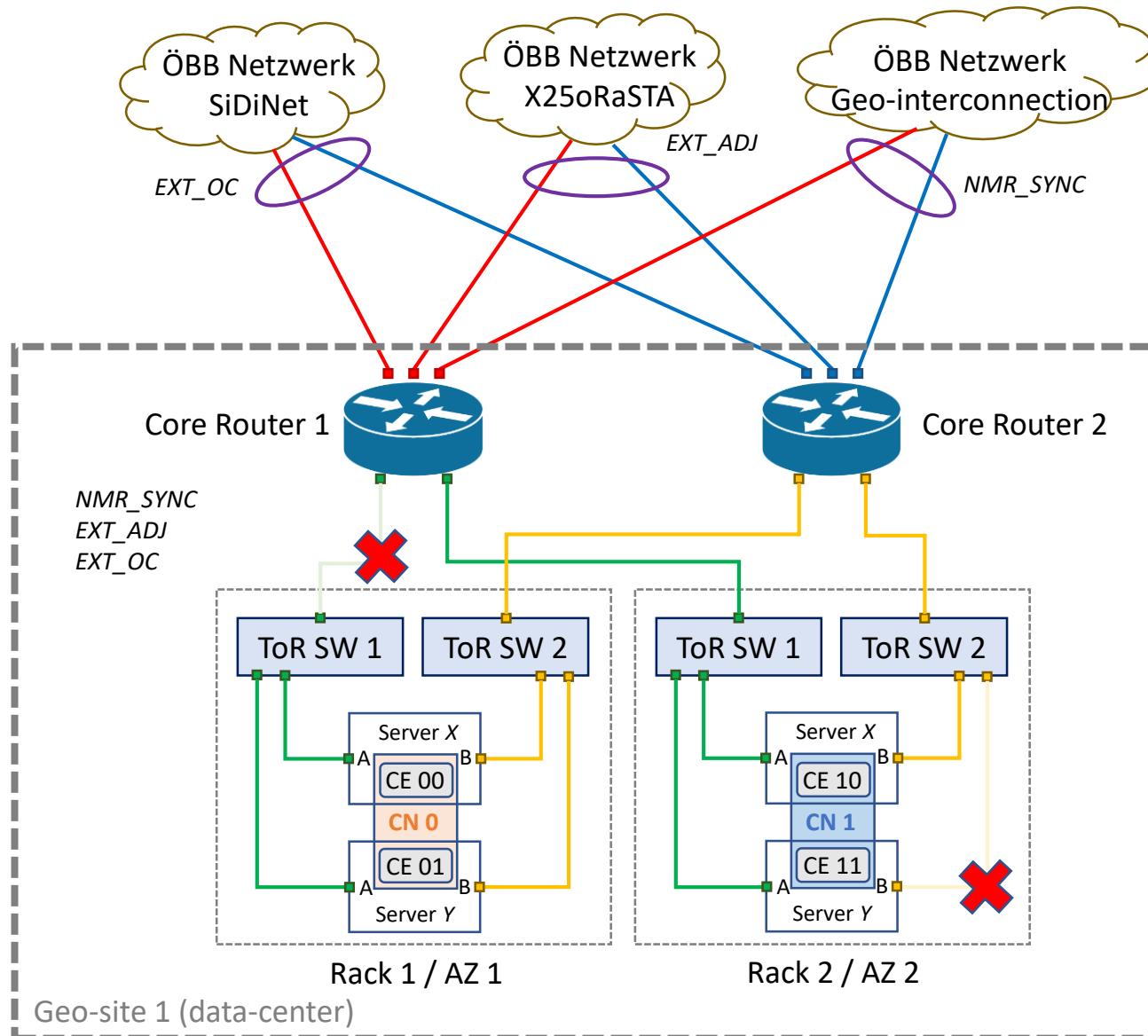
Mixed failures – between and
inside racks

Double failure between and inside racks



- **CN effect:** Only one physical channel but without effect on CN
- **CG effect:** CNO continues operation via “yellow” channel.
- **External networks:** No effect, full physical redundancy continues
- **No risk of split-brain**

Double failure between and inside racks



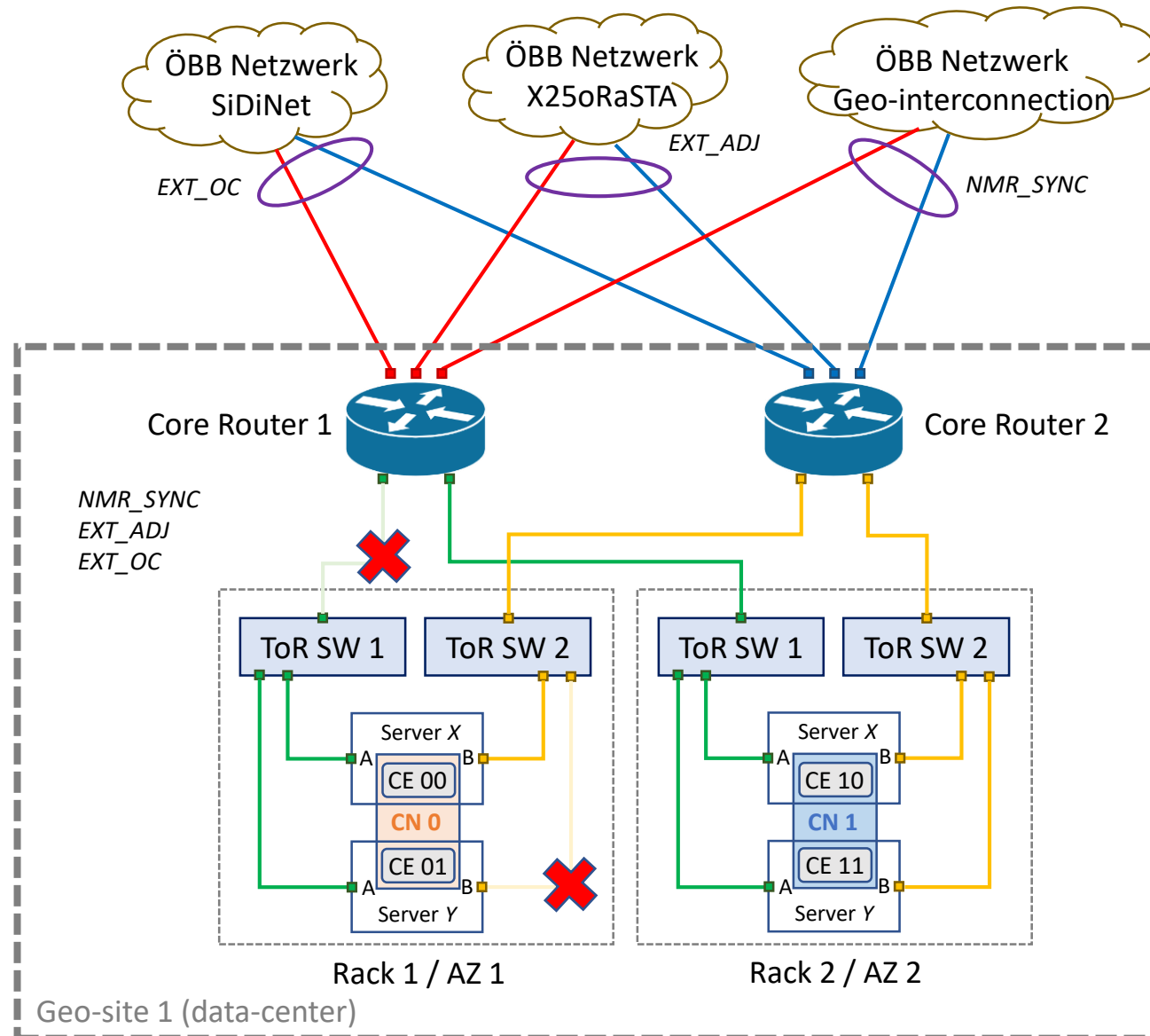
4x2oo2

- **CN effect:** Only one physical channel but without effect on CN
- **CG effect:**
 - **CN0 is Master:** CN1 will shutdown, CN2 and CN3 operates normally
 - **CN0 and CN1 are Slaves:** no effect
 - **CN1 is Master:** CN1 will shutdown, CN2 and CN3 operates normally
- **External networks:** no impact
- **No risk of split-brain**

2x2oo2

- **CN effect:** Only one physical channel but without effect on CN
- **CG effect:**
 - **CN0 is Master:** CN0 remains master. **CN1 - ???**
 - **CN0 is slave (CN1 master):** **???**
- **External networks:** Depends on the configuration, can be no impact or full interruption

Double failure between and inside racks



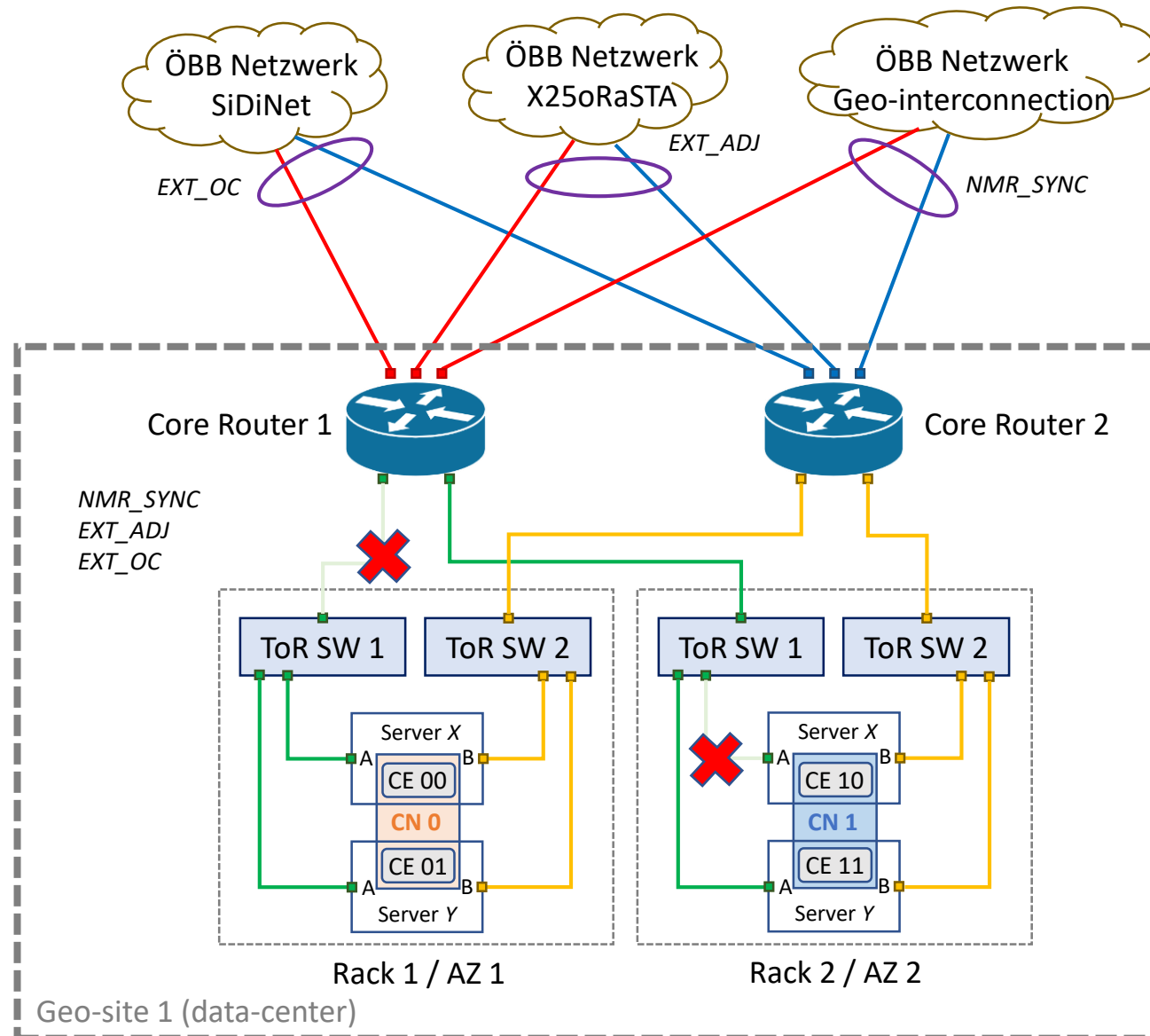
4x2oo2

- **CN effect:** Only one physical channel but without effect on CN
- **CG effect:**
 - CN0 is Master: No Master Tick msg received by other CNs – assume master is lost, starts MEP. MTP returns “Not okay” because no token sent. Slaves query MCI (or next check). Eventually will timeout (if MCI returns “not okay”) and all CNs go to recovery and elect new master afterwards without considering CN0.
 - New master eventually elected. Leader CE of CN0 receives master query – assumes network issues (CNs have state of the master as unknown) and goes panic.
 - CN0 is Slave: Goes to panic because Master tick message not received by both CEs.
- **External networks:**
 - No impact, CG will eventually vote new master. Old master will go to recovery.

2x2oo2

- **CG effect (Similar as in 4x2oo2):**
 - CN0 master: CN1 does not see master. MEP starts, MTP returns “N_OK”, MCI queried – if “N_OK” it will timeout and goes to the recovery and will eventually become new master.

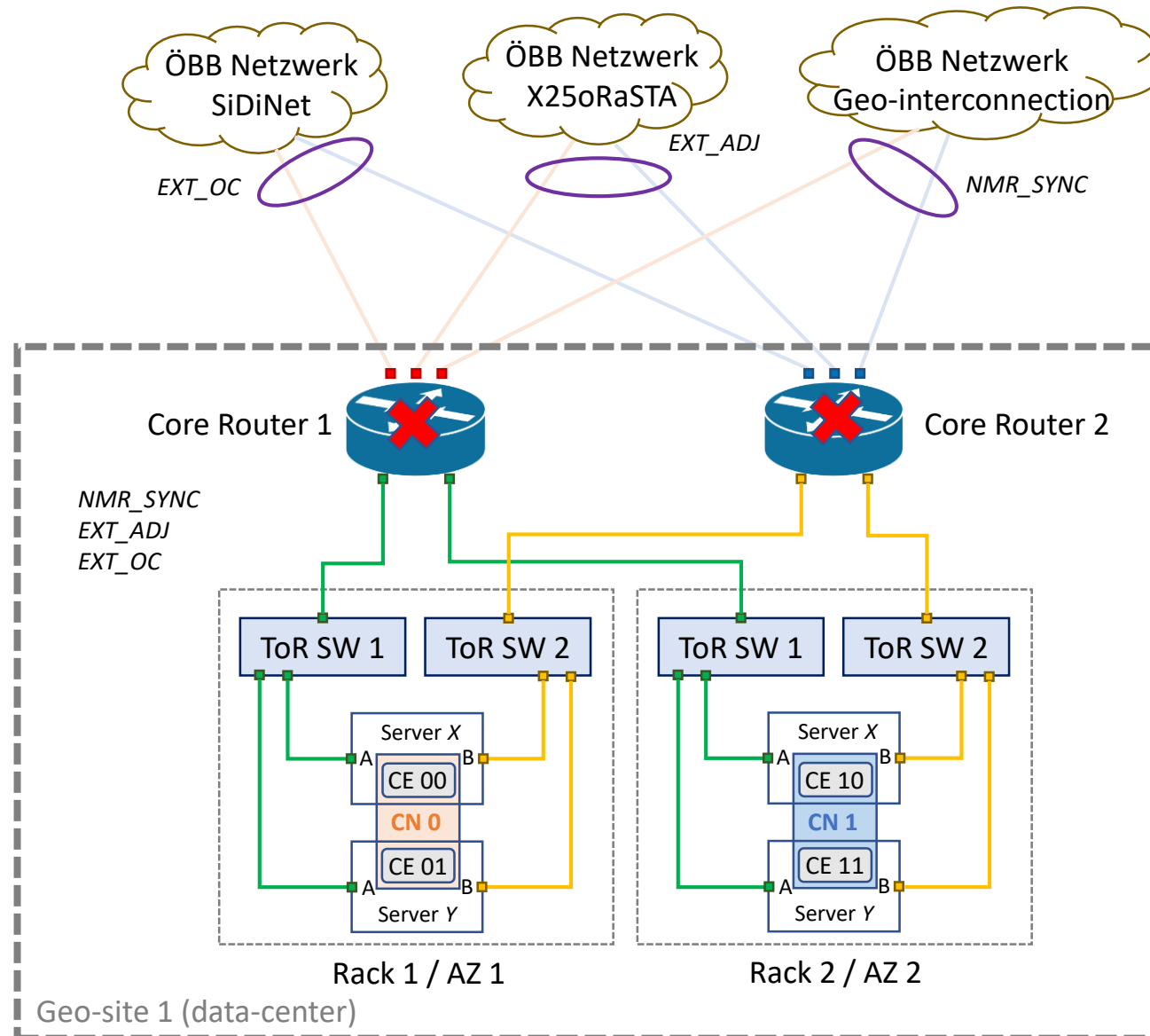
Double failure between and inside racks



- **CN effect:** Only one physical channel but without effect on CN
- **CG effect:** CN0 continues operation via “yellow” channel.
- **External networks:** No effect but only blue channel operational
- **No risk of split-brain**

Failures of core routers

Double failure of core routers



2x2oo2:

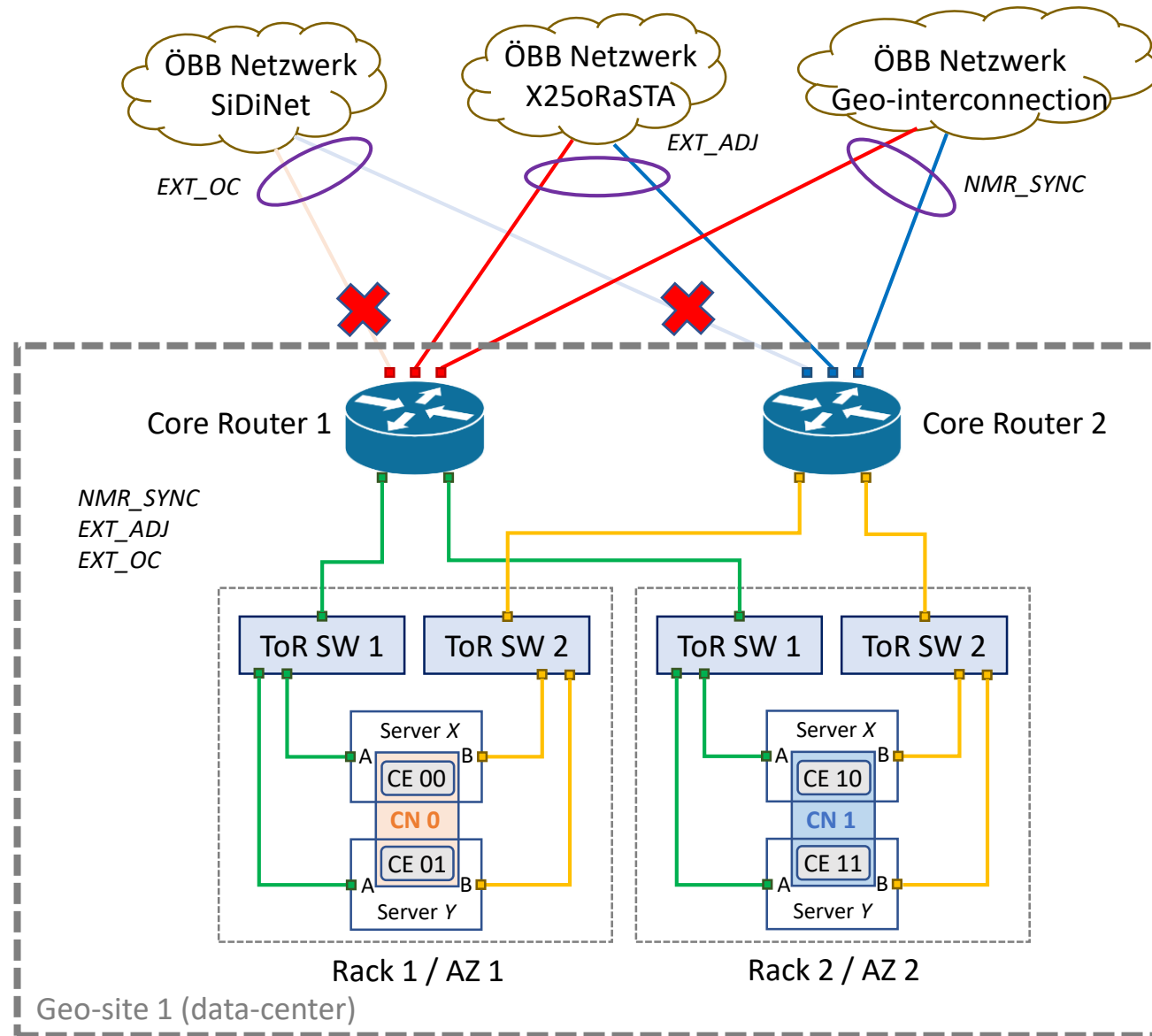
- **CN effect:** Both CNs are operational
- **CG effect (default):**
 - CNs cannot see each other
 - Master CN – remains master
 - Slave CN – Starts MEP
 - MTP cannot decide
 - MCI queried
- **External networks:** No communication with any external network
- **Split-brain could happen but no communication with external networks**
- **Availability issue – the entire VIL is isolated. If the connection re-establish and there were 2 masters, one will panic. No hazard because no change in the state**

4x2oo2:

- **CN effect:** Both CNs are operational
- **CG effect (default):**
 - CNs in the other geo-location cannot see CN in this location (state UNKNOWN) and vice versa.
 - Affected location has slaves – operation continues in the other geo-site. Slaves in affected location trigger MEP
 - MTP cannot decide
 - MCI queried
 - Affected location with the Master CN:
 - The other geo-loc will start MEP
 - MTP cannot decide
 - MCI queried
- **External networks:**
 - If site with slaves down – no impact
 - If site with the master down – until new master is elected, messages queued and processed after is elected
- **Split-brain could happen but no communication with external networks**
 - **If geo-site 2 elects also master, then safety hazard when re-integrating because site 2 could have advance and CN might be elected**
- **Availability issue – if site with the master is down**

Failures in external networks

Double failure in SiDiNet



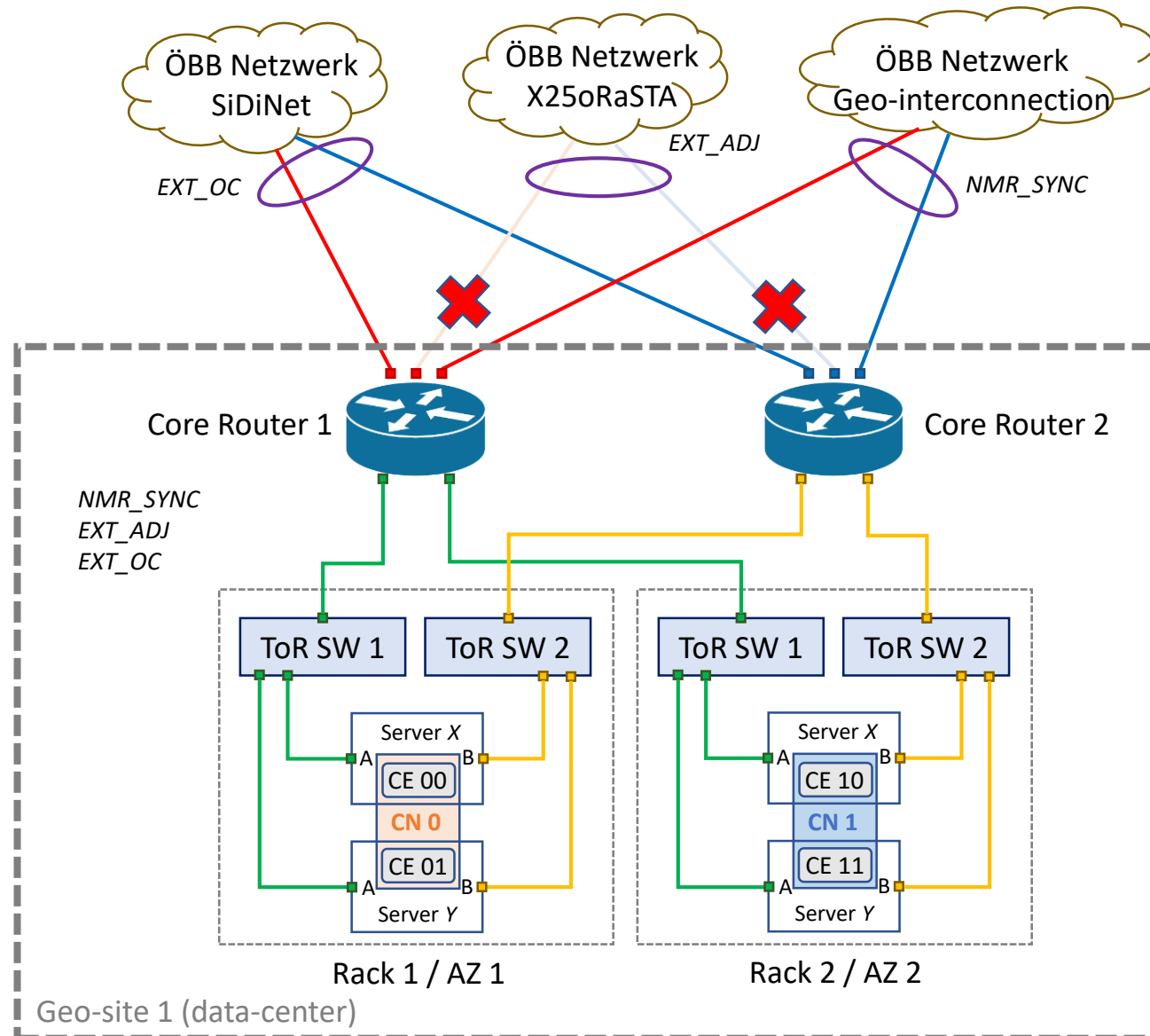
2x2oo2:

- **CN effect:** All 2 CNs are operational
- **CG effect (default):** no effect
- **External networks:** Loss of connection with field elements
- **No risk of split brain**
- **Availability issue – no communication with field elements**

4x2oo2:

- **CN effect:** All 4 CNs are operational
- **CG effect (default):** no effect
- **External networks:** The other geo-site continues communication with field elements
- **No risk of split brain**
- **No availability issue**

Double failure in X25oRaSTA network



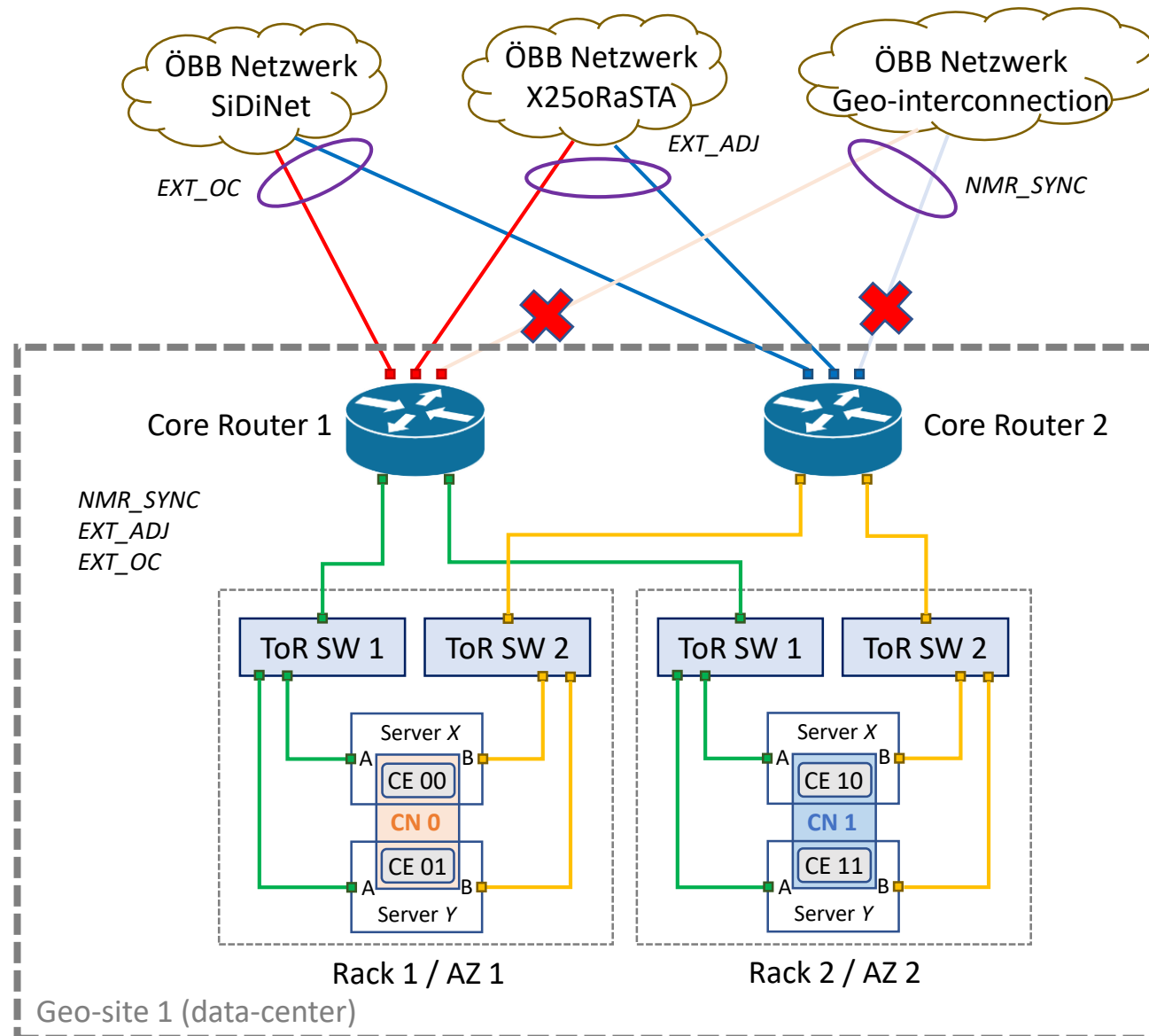
2x2oo2:

- **CN effect:** All 2 CNs are operational
- **CG effect (default):** no effect
- **External networks:** Loss of connection with adjacent systems
- **No risk of split brain**
- **Availability issue – no communication with adjacent systems**

4x2oo2:

- **CN effect:** All 4 CNs are operational
- **CG effect (default):** no effect
- **External networks:** The other geo-site continues communication with adjacent systems
- **No risk of split brain**
- **No availability issue**

Double failure in geo-interconnect network



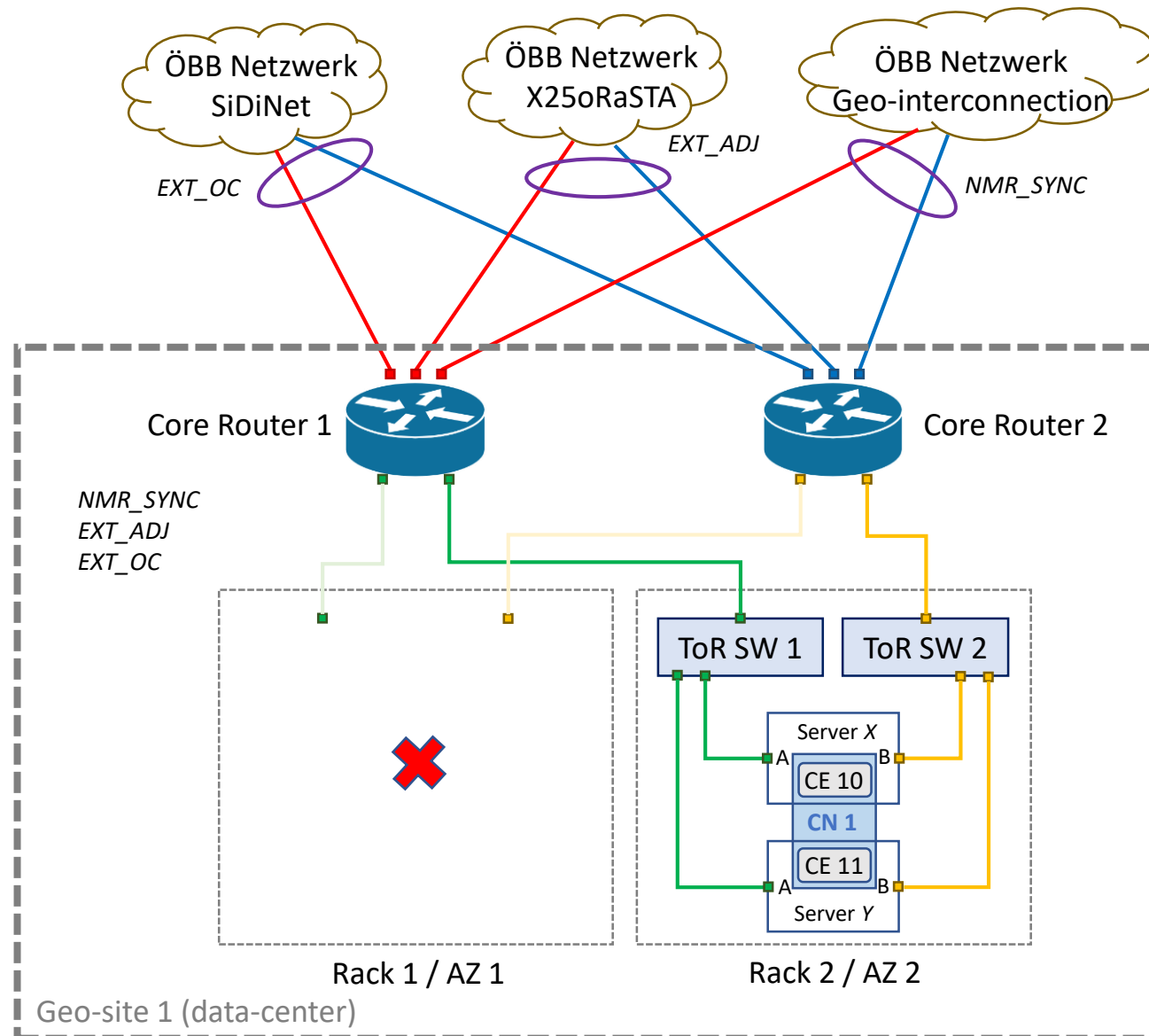
2x2oo2: not applicable

4x2oo2:

- **CN effect:** All 4 CNs are operational
- **CG effect (default):**
 - CNs in different geo-location cannot see each other
 - Location with Master CN – remains master
 - Location with Slave CN – Starts MEP
 - MTP cannot decide
 - MCI queried
- **External networks:** Site with master CN continues communication with all EXT. Site with slaves will not be operational
- **No risk of split brain**
 - No safety hazard because site with master continues and site with slaves is unable to elect the master
 - Risk only when reintegrating and
- **No availability issue**
 - Site with master is connected to all external networks

Power outages

Power outage in a single rack (AZ)



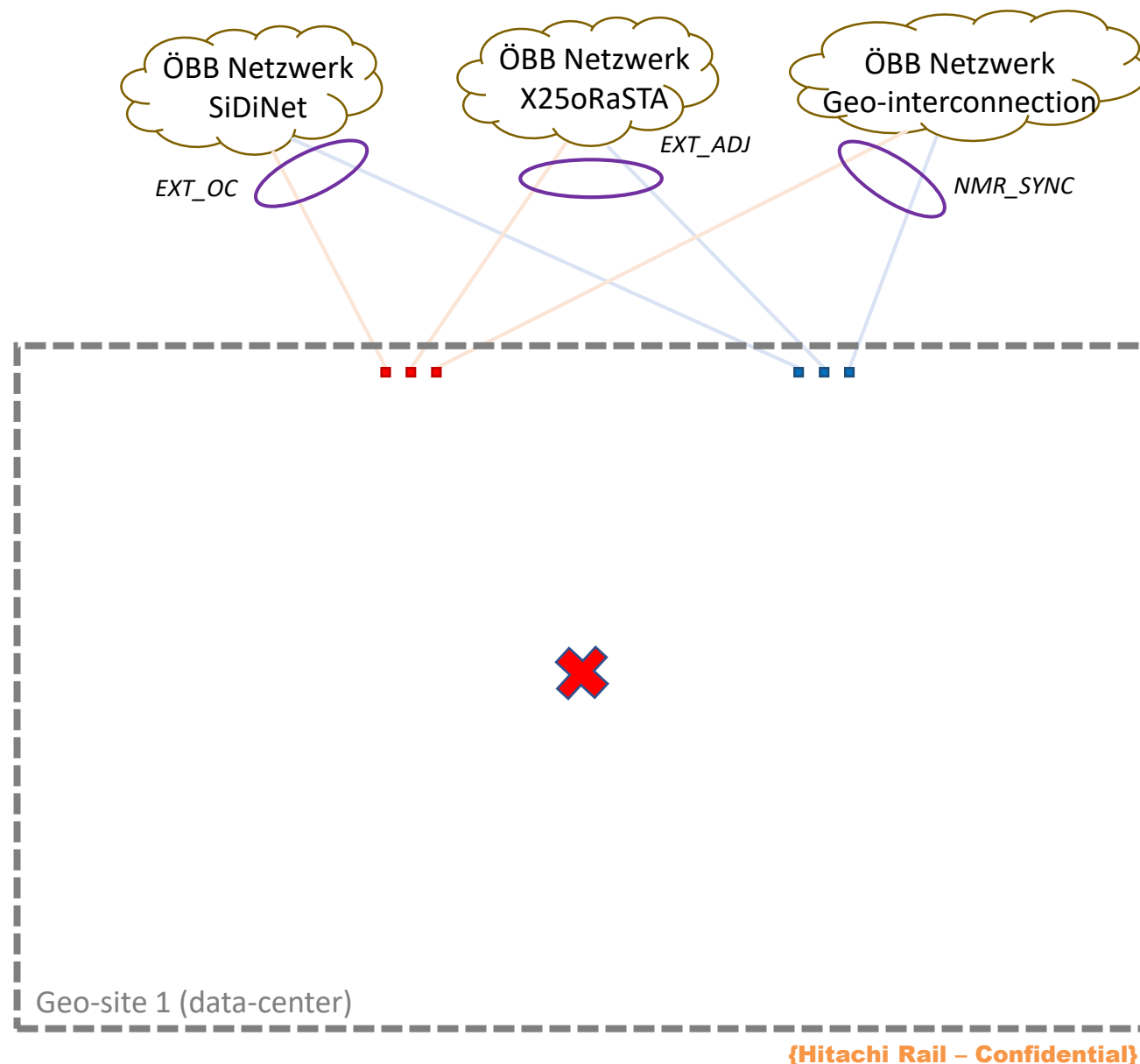
Master CN:

- **CN effect:** CN is dead
- **CG effect (default):**
 - Master election process (MEP) starts
 - MTP does not elect master
 - MCI queried
- **External networks:** No communication with any network
- **No risk of split-brain**
 - No safety hazard because CN0 that was master is down and neither communicates nor computes
- **Availability issue – other redundancies cannot elect master (unless other measures)**

Slave CN:

- **CN effect:** CN is dead
- **CG effect (default):**
 - The rest of CG continues normal operation
 - If MEP is triggered (in the future), no impact
- **External networks:** No effect
- **No risk of split-brain**

Power outage of the entire geo-site



2x2oo2:

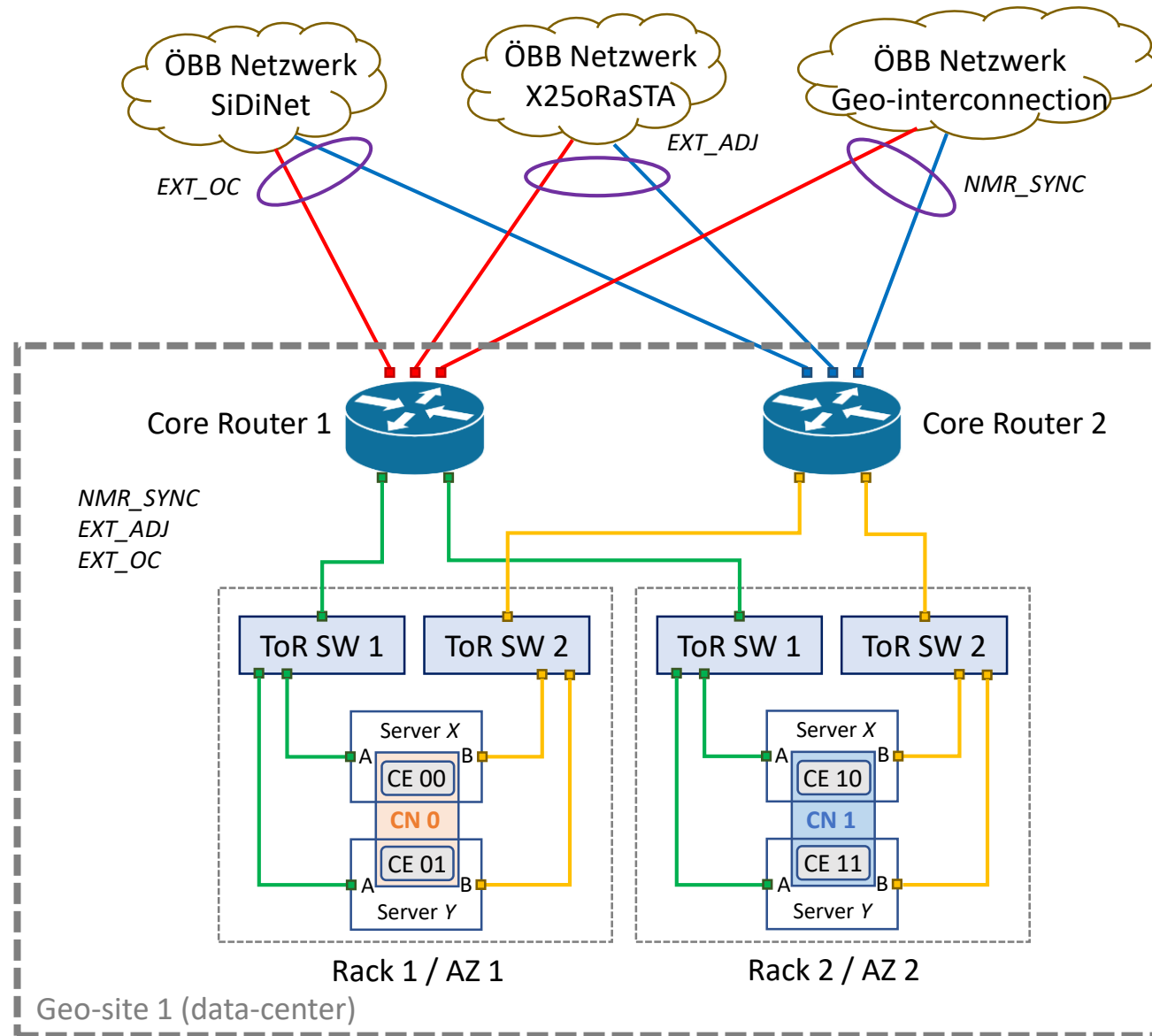
- The whole VIL is down, no operations
- No risk of split-brain
- **No risk of split brain**
- **Availability issue – the entire VIL is shutdown**

4x2oo2:

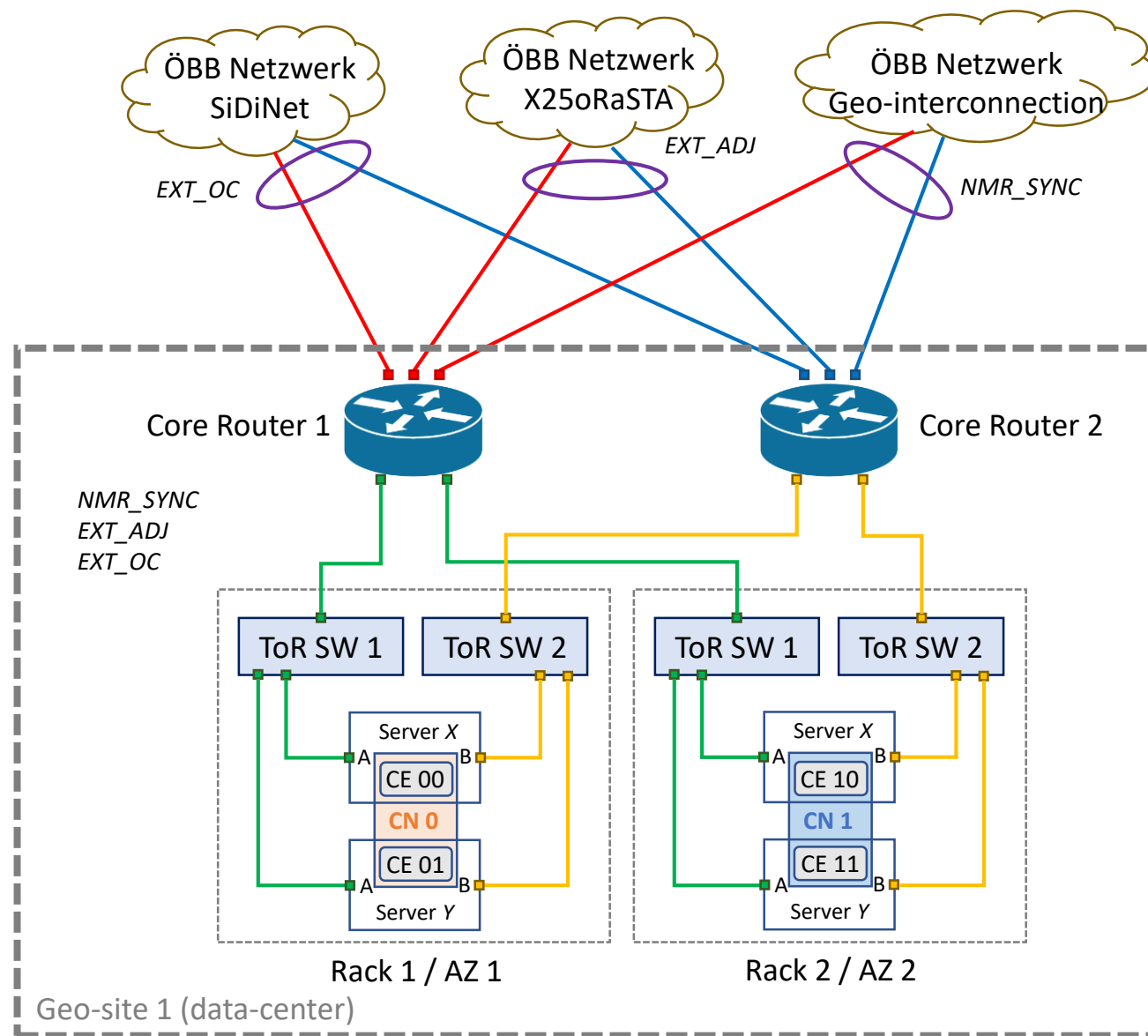
- **CN effect:** Both CNs are down
- **CG effect (default):**
 - CNs in the other geo-location cannot see CN in this location (state UNKNOWN)
 - Location with slaves down – operation continues in the other site
 - Location with the Master CN down:
 - The other geo-loc will start MEP
 - MTP cannot decide
 - MCI queried
- **External networks:**
 - If site with slaves down – no impact
 - If site with the master down – operation stops
- **No risk of split brain**
- **Availability issue – if site with the master is down**

Backup slides

Network topology for split-brain analysis



Network topology for split-brain analysis



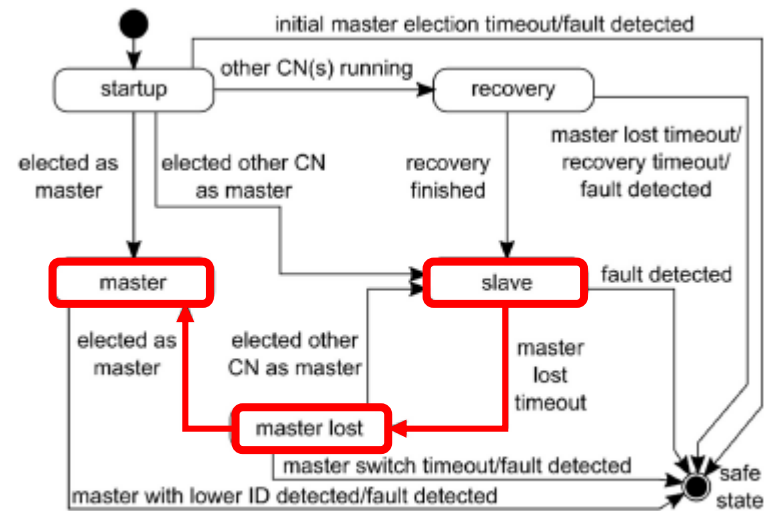


Figure 3 NMR state machine of a CN

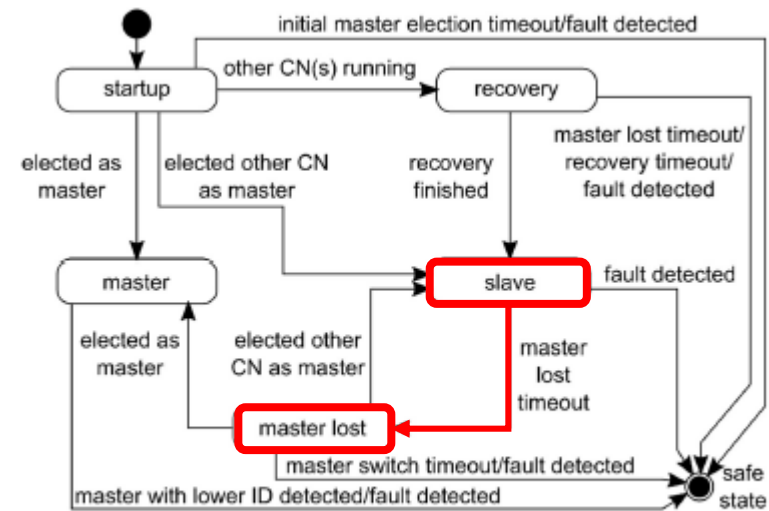


Figure 3 NMR state machine of a CN